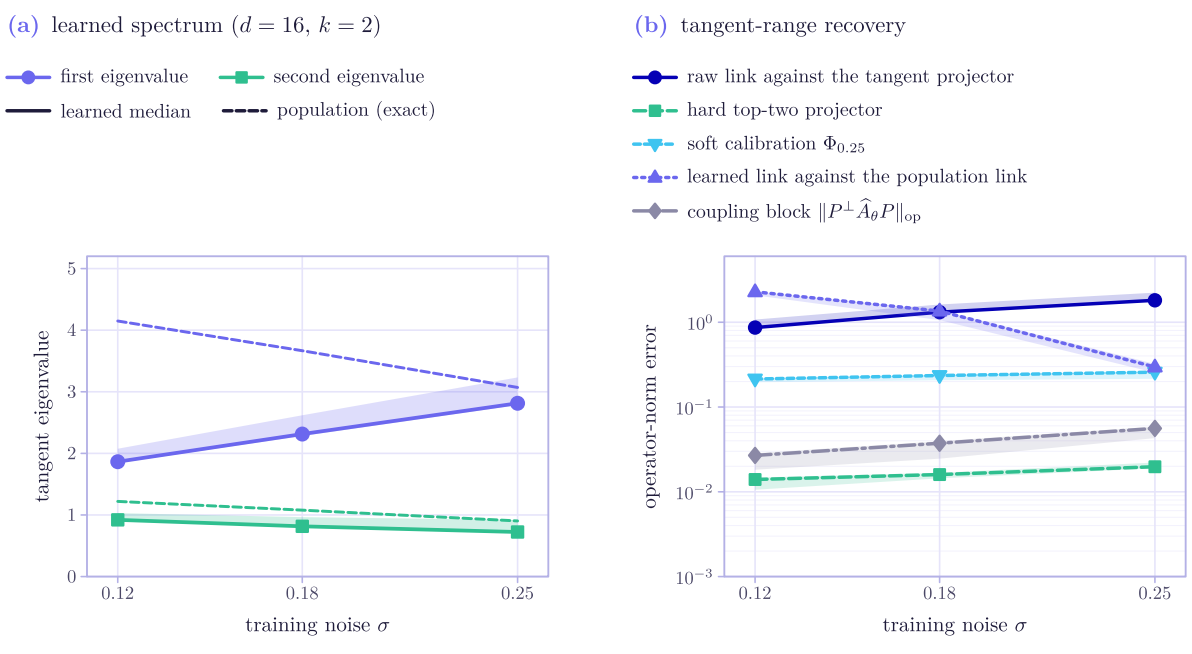

(a) learned spectrum ($d = 16$, $k = 2$)
first eigenvalue
second eigenvalue
learned median
population (exact)
tangent eigenvalue
0
1
2
3
4
5
0.12
0.18
0.25
training noise σ
(b) tangent-range recovery
raw link against the tangent projector
hard top-two projector
soft calibration $\Phi_{0.25}$
learned link against the population link
coupling block $\|P^{\perp}\widehat{A}_{\theta}P\|_{\mathrm{op}}$
operator-norm error
10^{0}
10^{-1}
10^{-2}
10^{-3}
0.12
0.18
0.25
training noise σ